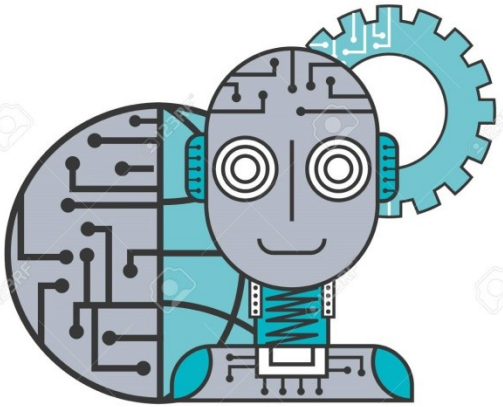
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Module 6CM521: Frontiers in AI and Data Science



Data patterns

Ioannis Tsioulis (MC)

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Slides acknowledgements:
Dr Hong Qing Yu

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Declare

This presentation is mostly based on the book: Data Mining, Concepts and Techniques, *Chapter 6*, Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods.

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Agenda

- **Basic Concepts**
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

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Frequent Pattern Analysis Example

TID	Milk	Bread	Butter	Beer
1	1	0	1	1
2	1	1	1	0
3	0	1	1	0
4	1	0	0	1
5	1	1	1	1

How do we identify these frequent patterns from dataset (big)?

How do we justify the pattern frequencies?

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What Is Frequent Pattern Analysis?

- Frequent pattern: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [SIGMOD'93] in the context of frequent itemsets and association rule mining

Mining Association Rules between Sets of Items in Large Databases

Rakesh Agrawal Tomasz Imielinski* Arun Swami

IBM Almaden Research Center
650 Harry Road, San Jose, CA 95120

Abstract

We are given a large database of customer transactions. Each transaction consists of items purchased by a customer in a visit. We present an efficient algorithm that generates all significant association rules between items in the database. The algorithm incorporates buffer management and novel estimation and pruning techniques. We also present results of applying this algorithm to sales data obtained from a large retailing company, which shows the effectiveness of the algorithm.

Several organizations have collected massive amounts of such data. These data sets are usually stored on tertiary storage and are very slowly migrating to database systems. One of the main reasons for the limited success of database systems in this area is that current database systems do not provide necessary functionality for a user interested in taking advantage of this information.

This paper introduces the problem of "mining" a large collection of basket data type transactions for associa-

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Motivations of Frequent Pattern Analysis

Finding inherent regularities in data

- What products were often purchased together?— Beer and diapers?!
- What are the subsequent purchases after buying a PC?
- What kinds of DNA are sensitive to this new drug?
- Can we automatically classify web documents?

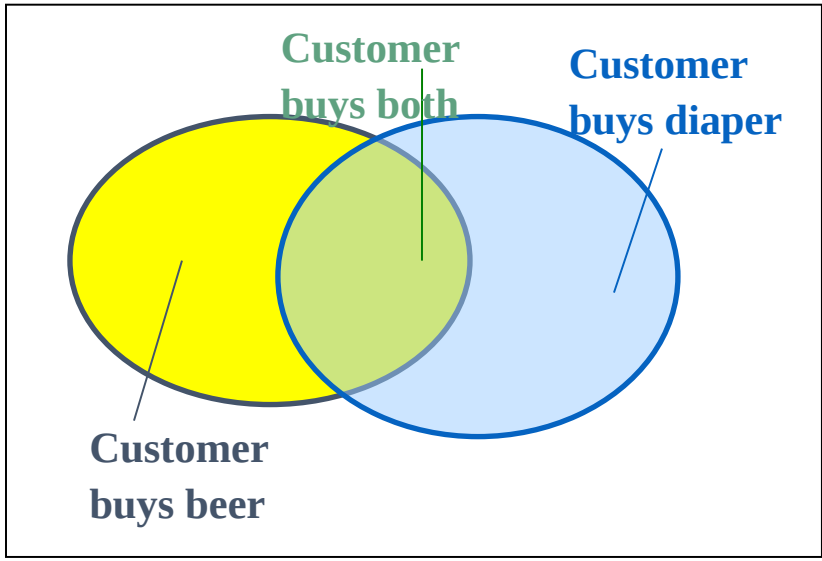
Applications

- Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

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Basic Concepts: Frequent Patterns

Items bought
Beer, Nuts, Diaper
Beer, Coffee, Diaper
Beer, Diaper, Eggs
Nuts, Eggs, Milk
Nuts, Coffee, Diaper, Eggs, Milk



Itemset(Dataset):
A set of one or more items

k-itemset $X = \{x_1, \ldots, x_k\}$

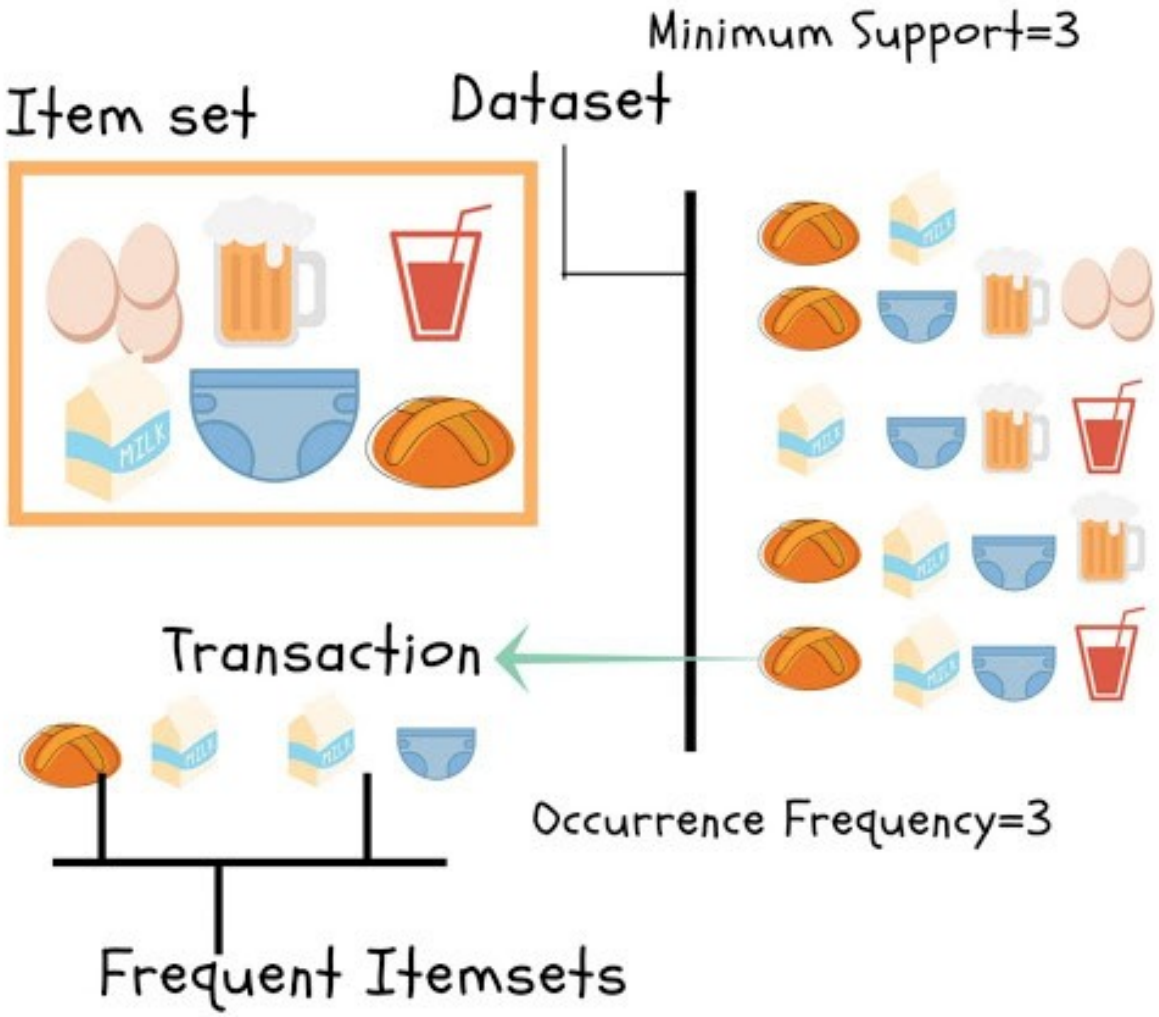
(absolute) support, or, *support count* of X:
Frequency or occurrence of an itemset X

(relative) support, *s*, is the fraction of transactions that contains X (i.e., the probability that a transaction contains X)

An itemset X is *frequent* if X's support is no less than a *minsup* threshold

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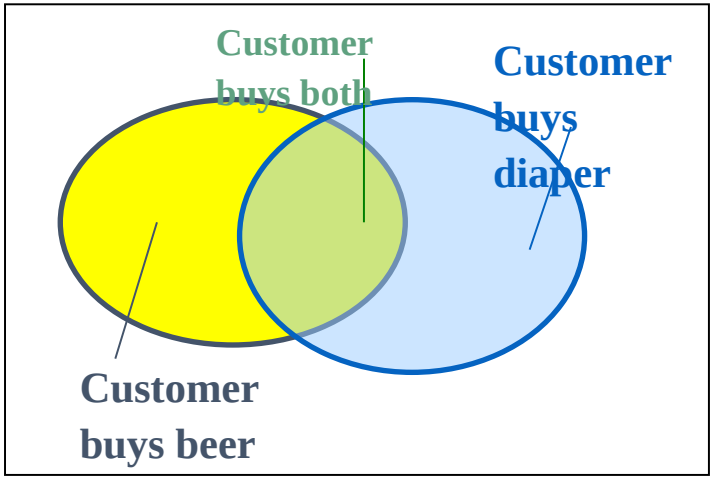
Frequent Patterns Example



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Basic Concepts: Association Rules

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



Find all the rules $X \rightarrow Y$ with minimum support and confidence

- Relative support, *s*, probability that a transaction contains $X \cup Y$
- confidence, *c*, conditional probability that a transaction having X also contains *Y*

Let minsup = 50%, minconf = 50%

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer, Diaper}:3

- Association rules: (many more!)
  - *Beer → Diaper* (60%, 100%)
  - *Diaper → Beer* (60%, 75%)

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Association Rule Example

Basket	Product 1	Product 2	Product 3
1	Milk	Cheese	
2	Milk	Apples	Cheese
3	Apples	Banana	
4	Milk	Cheese	
5	Apples	Banana	
6	Milk	Cheese	Banana
7	Milk	Cheese	
8	Cheese	Banana	
9	Cheese	Milk	

What is the support value milk + cheese?

$$\text{Support} = \frac{(A + B)}{\text{Total}}$$

$$\text{Support for Basket 1} = \frac{(\text{Milk} + \text{Cheese})}{\text{Total}} = \frac{6}{9} = .6666667$$

What is the confidence value?

$$\text{Confidence} = \frac{(A + B)}{A}$$

$$\text{Confidence for Basket 1} = \frac{(\text{Milk} + \text{Cheese})}{\text{Milk}} = \frac{6}{6} = 1.000$$

(!new!) The lift or lift ratio measure of confidence to expected confidence.

Expected confidence is the confidence divided by the frequency of B. The Lift tells us how much better a rule is at predicting the result than just assuming the result in the first place. Greater lift values indicate stronger associations.

Lift ration

$$\text{Lift} = \left(\frac{\left(\frac{(A + B)}{A}\right)}{\left(\frac{B}{\text{Total}}\right)}\right)$$

$$\text{Lift for Basket 1} = \left(\frac{\left(\frac{(\text{Milk} + \text{Cheese})}{\text{Milk}}\right)}{\left(\frac{\text{Cheese}}{\text{Total}}\right)}\right) = \left(\frac{\left(\frac{6}{6}\right)}{\left(\frac{7}{9}\right)}\right) = \left(\frac{1}{.7777778}\right) = 1.2857$$

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Sub & Super Patterns

Sub-pattern (proper subset)

We call any proper subset of a pattern, a sub-pattern. Here is an example,

If a pattern X contains, 3 items: X={a, b, c}, any proper subsets, such as sets containing a, or b, or c, or ab, or ac, or bc, are all sub patterns.

Now, suppose we have a pattern with 100 items. How many sub-patterns can we find for this pattern?

A lot, it is more than 10 to the power of 30.

Super-pattern (proper superset)

A super pattern of a pattern is a proper super set of the pattern.

Here are some examples of the pattern containing the 3 items: a, b, c

abcd, abce, abcde, …

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Close Pattern

A **closed pattern** is a frequent pattern. So it meets the minimum support criteria. In addition to that, all super-patterns of a closed pattern are less frequent than the closed pattern.

Formally, this is defined as:

CP is a **Closed Pattern** if

- CP is a **frequent pattern** and
- All super-patterns of CP are less frequent than CP
  - For all super-patterns S of CP,
    Supp(S) < Supp(CP)

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Close Pattern example

Suppose, the minimum support count is 2.

For the first example, suppose there are a total of 3 items: a, b, c.

Suppose a pattern ab has support count of 2 and a pattern abc has support count of 2.

Is the pattern ab is a closed pattern? No

Pattern ab is a frequent pattern, but it has a super-pattern that is NOT less frequent than ab.

For the second example,

suppose there are a total of 3 items: x, y, z.

suppose a pattern xy has support count of 3 and a pattern xyz has support count of 2.

Is the pattern xy is a closed pattern? Yes

Pattern xy is a frequent pattern and also the only super-pattern xyz is less frequent than xy.

Therefore, xy is a closed pattern.

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Max Pattern

A **max pattern** is

1. a frequent pattern. So it also meets the minimum support criteria like closed pattern
2. In addition, but unlike closed pattern, all super-patterns of a max pattern are NOT frequent patterns.

This is formally written as

MP is a **Max Pattern** if

- MP is a frequent pattern and
- All super-patterns of MP are not frequent
  - For all super-pattern SP of MP, $Supp(SP) < Supp_{Min}$

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Max Pattern examples

Suppose, the minimum support count is 2.

Like before, for the first example, suppose there are a total of 3 items: a, b, c.

Suppose a pattern ab has support count of 3 and a pattern abc has support count of 2.

Is the pattern ab is a max pattern? No

Pattern ab is a frequent pattern, but it has a super-pattern that is a frequent pattern as well. So, pattern ab is NOT a max pattern.

For the second example,

suppose there are a total of 3 items: x, y, z.

Suppose a pattern xy has support count of 3 and a pattern xyz has support count of 1.

Is the pattern xy is a max pattern? Yes

Pattern xy is a frequent pattern and also the only super-pattern xyz is NOT a frequent pattern. Therefore, xy is a max pattern.

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Closed Patterns (CP) and Max-Patterns (MP)

A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \ldots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \ldots + \binom{100}{100} = 2^{100} - 1 = 1.27*10^{30}$ sub-patterns!

Solution: *Mine closed patterns and max-patterns instead*

- An itemset X is closed if X is *frequent* and there exists *no super-pattern* Y ⊃ X, *with the same support* as X
- An itemset X is a max-pattern if X is frequent and there exists no frequent super-pattern Y ⊃ X

Closed pattern is a lossless compression of freq. patterns

- Reducing the # of patterns and rules

Max-pattern is a lossy compression

- We only know max-pattern is freq. pattern but not sure the frequent support numbers.

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Computational Complexity of Frequent Itemset Mining

How many itemsets are potentially to be generated in the worst case?

- The number of frequent itemsets to be generated is sensitive to the minsup threshold
- When minsup is low, there exist potentially an exponential number of frequent itemsets
- The worst case: M^N where M: # distinct items, and N: max length of transactions

The worst case complexity vs. the expected probability

- Ex. Suppose TESCO has 10^4 kinds of products
  - The chance to pick up one product 10^{-4}
  - The chance to pick up a particular set of 10 products: ~10^{-40}
  - What is the chance this particular set of 10 products to be frequent 10^3 times in 10^9 transactions?

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Agenda

- Basic Concepts
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The Downward Closure Property and Scalable Mining Methods

Three major approaches

- Apriori (Agrawal & Srikant@VLDB'94)
- Freq. pattern growth (FPgrowth—Han, Pei & Yin @SIGMOD'00)
- Vertical data format approach (Charm—Zaki & Hsiao @SDM'02)

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Apriori: A Candidate Generation & Test Approach

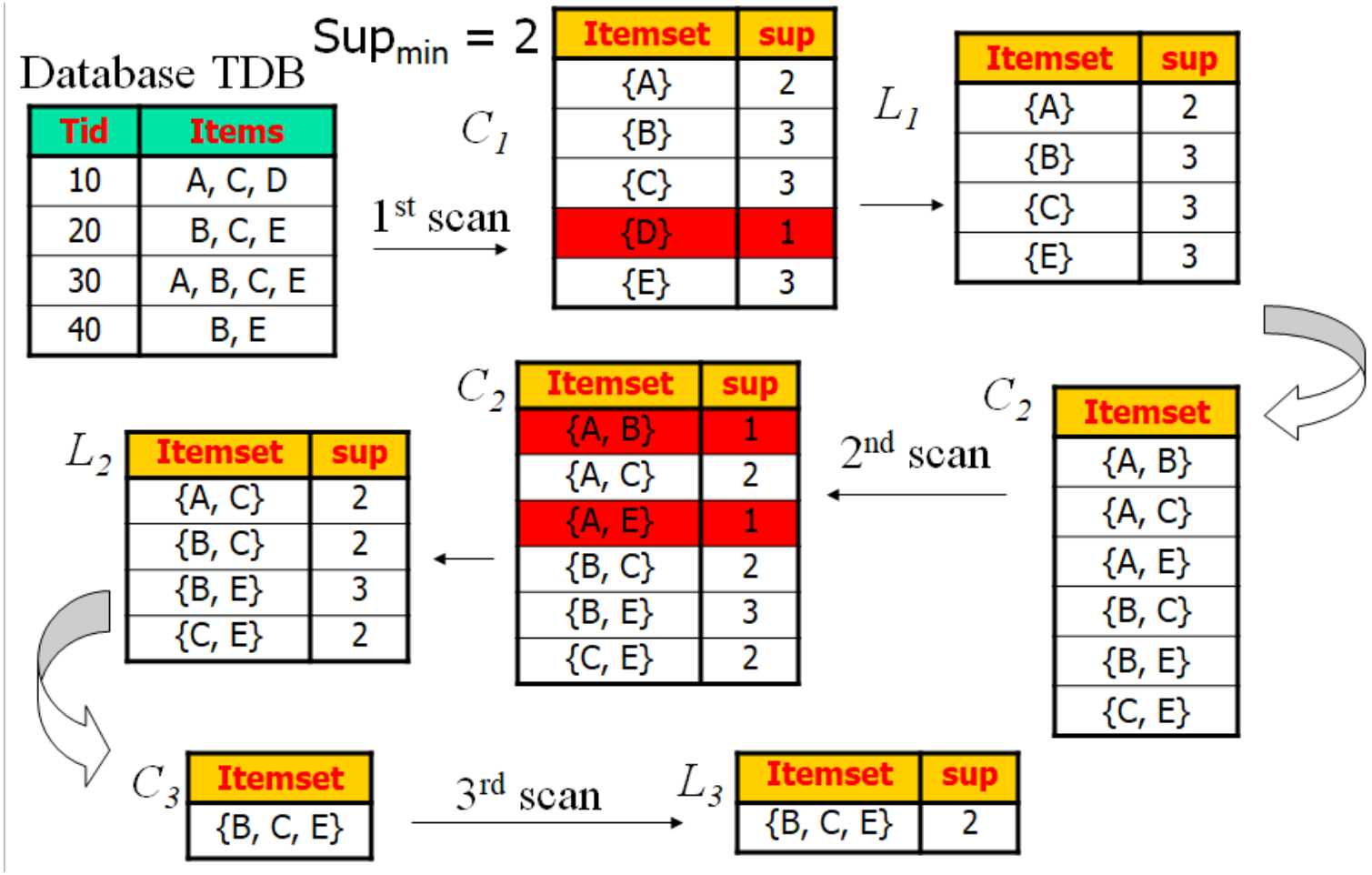
Method:

- Initially, scan DB once to get frequent 1-itemset
- Generate length (k+1) candidate itemsets from length k frequent itemsets
- Test the candidates against DB
- Terminate when no frequent or candidate set can be generated

Apriori pruning principle: If there is any itemset which is infrequent, its superset should not be generated/tested! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)

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The Apriori Algorithm—An Example



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The Apriori Algorithm - Pseudo-Code

C_k: Candidate itemset of size k

L_k : frequent itemset of size k

L_1 = {frequent items};

for (k = 1; L_k !=∅; k++) **do begin**

  C_{k+1} = candidates generated from L_k;

  for each transaction t in database do

    increment the count of all candidates in C_{k+1}

    that are contained in t

  L_{k+1} = candidates in C_{k+1} with min_support

  end

return $\cup_k L_k$;

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Further Improvement of the Apriori Method

Major computational challenges

- Multiple scans of transaction database
- Huge number of candidates
- Tedious workload of support counting for candidates

Improving Apriori: general ideas

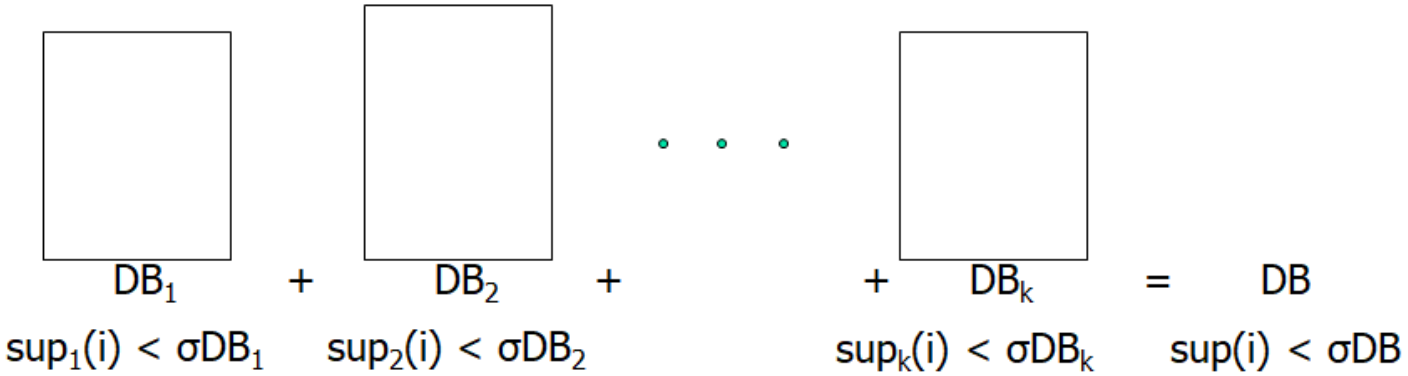
- Reduce passes of transaction database scans
- Shrink number of candidates
- Facilitate support counting of candidates

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Partition: Scan Database Only Twice

Any itemset that is potentially frequent in DB must be frequent in at least one of the partitions of DB

- Scan 1: partition database and find local frequent patterns
- Scan 2: consolidate global frequent patterns



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DHP: Reduce the Number of Candidates

A *k*-itemset whose corresponding hashing bucket count is below the threshold cannot be frequent

- Candidates: a, b, c, d, e
- Hash entries
  - {ab, ad, ae}
  - {bd, be, de}
  - …
- Frequent 1-itemset: a, b, d, e
- ab is not a candidate 2-itemset if the sum of count of {ab, ad, ae} is below support threshold

Minsup=100

count	itemset
35	{ab, ad, ae}
88	{bd, be, de}
. . .	. . .
102	{yz, qs, wt}

J. Park, M. Chen, and P. Yu. An effective hash-based algorithm for mining association rules. *SIGMOD'95*

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More ideas of improvement

H. Toivonen. Sampling large databases for association rules. In *VLDB'96*

S. Brin R. Motwani, J. Ullman, and S. Tsur. Dynamic itemset counting and implication rules for market basket data. *SIGMOD'97*

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Extension of Pattern Growth Mining Methodology

Mining closed frequent itemsets and max-patterns

- CLOSET (DMKD'00), FPclose, and FPMax (Grahne & Zhu, Fimi'03)

Mining sequential patterns

- PrefixSpan (ICDE'01), CloSpan (SDM'03), BIDE (ICDE'04)

Mining graph patterns

- gSpan (ICDM'02), CloseGraph (KDD'03)

Constraint-based mining of frequent patterns

- Convertible constraints (ICDE'01), gPrune (PAKDD'03)

Computing iceberg data cubes with complex measures

- H-tree, H-cubing, and Star-cubing (SIGMOD'01, VLDB'03)

Pattern-growth-based Clustering

- MaPle (Pei, et al., ICDM'03)

Pattern-Growth-Based Classification

- Mining frequent and discriminative patterns (Cheng, et al, ICDE'07)

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ECLAT: Mining by Exploring Vertical Data Format

Vertical format: t(AB) = {T_{11}, T_{25}, …}

- tid-list: list of trans.-ids containing an itemset

Deriving frequent patterns based on vertical intersections

- t(X) = t(Y): X and Y always happen together
- t(X) ⊂ t(Y): transaction having X always has Y

Using diffset to accelerate mining

- Only keep track of differences of tids
- t(X) = {T_1, T_2, T_3}, t(XY) = {T_1, T_3}
- Diffset (XY, X) = {T_2}

Eclat (Zaki et al. @KDD'97)

Mining Closed patterns using vertical format: CHARM (Zaki & Hsiao@SDM'02)

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Horizontal Data Layout

TID	Items
1	A,B,E
2	B,C,D
3	C,E
4	A,C,D
5	A,B,C,D
6	A,E
7	A,B
8	A,B,C
9	A,C,D
10	B

Vertical Data Layout

A	B	C	D	E
1	1	2	2	1
4	2	3	4	3
5	5	4	5	6
6	7	8	9	
7	8	9		
8	10			
9				

TID-list

A
1
4
5
6
7
8
9

$\wedge$

B
1
2
5
7
8
10

$\rightarrow$

AB
1
5
7
8

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Summary

Basic concepts: association rules, support-confident framework, closed and max-patterns, Lift association measurement method

Scalable frequent pattern mining methods

- Apriori
- Projection-based
- Vertical format approach

Next:

Lecture: Decision Tree (Classification problem)

Practical: Decision Tree with Python

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Ref: Basic Concepts of Frequent Pattern Mining

(Association Rules) R. Agrawal, T. Imielinski, and A. Swami. Mining association rules between sets of items in large databases. SIGMOD'93

(Max-pattern) R. J. Bayardo. Efficiently mining long patterns from databases. SIGMOD'98

(Closed-pattern) N. Pasquier, Y. Bastide, R. Taouil, and L. Lakhal. Discovering frequent closed itemsets for association rules. ICDT'99

(Sequential pattern) R. Agrawal and R. Srikant. Mining sequential patterns. ICDE'95

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S. Sarawagi, S. Thomas, and R. Agrawal. Integrating association rule mining with relational database systems: Alternatives and implications. SIGMOD'98

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Ref: Mining Correlations and Interesting Rules

S. Brin, R. Motwani, and C. Silverstein. Beyond market basket: Generalizing association rules to correlations. SIGMOD'97.

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